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question one

Characteristic	Requirement detail	
Optimal Temperature Range	Sunflowers thrive in warm weather. They grow best in high summer temperatures and are sensitive to frost.	21°C – 26°C (70°F – 80°F)
Optimal Relative Humidity Range	They prefer moderate to dry ambient air. High humidity combined with heat increases the risk of leaf rust and fungal infections.	40% – 50% RH
Recommended Soil Type	They need nutrient-rich soil that accommodates their long taproots. While adaptable, they hate stagnant water.	Deep, well-draining loamy soil
Optimal Soil Moisture Content	They are relatively drought-tolerant once established but need consistent moisture during germination and budding.	35% – 50% Volumetric Water Content (VWC)
Optimal Soil pH Range	They prefer a neutral environment but can tolerate slightly acidic to slightly alkaline conditions.	6.0 – 7.5 pH (6.5 – 7.0 is ideal)
. Hours of Sunlight Exposure	True to their name, they are heliotropic and demand full, unobstructed sun to support heavy flower heads.	6 – 8 hours minimum (Up to 10 hours of direct sunlight is preferred)

2. Hardware Component List

Core Computational & Display Modules

- **ESP32S DevKIT Wi-Fi + BLE Module (30-Pin):** The central microcontroller brain that processes sensor readings and sends data over the cloud.
- **1.3" White IIC 128X64 OLED LCD Display:** The on-device screen used to output local real-time metrics to the user.

Sensor Network

- **DHT22 (AM2302) Sensor:** Captures ambient **Air Temperature** and **Relative Humidity**.
- **MQ-5 Gas Sensor:** Measures targeted concentrations of **LPG, methane, butane, and propane**.
- **Soil Moisture Sensor (e.g., Capacitive v1.2):** Required to fulfill metric *1.d* (soil moisture tracking).
- **Soil pH Sensor Probe:** Required to fulfill metric *1.e* (soil pH tracking).
- **BH1750 Light Sensor (or Photoresistor):** Required to track ambient light levels over time to calculate total daily **Sunlight Exposure** hours (*1.f*).

Power, Control, & Prototyping Tools

- **5V 1-Channel Low-Level Trigger Relay Module:** Acts as an automated switch to turn external hardware on/off (e.g., activating a water pump if soil moisture drops below 40%).
- **Solderless Breadboard (Full-size):** Temporary testing base to link all components without permanent soldering.
- **Premium Jumper Wires (Male-to-Male, Male-to-Female):** Interconnecting wires to establish electrical connections between the ESP32 and the sensors.
- **Micro-USB or Type-C Cable:** For flashing code from your computer and delivering 5V power to the development board.

3. 1.3" White IIC 128X64 OLED LCD Display

- [OLED Display Module SH1106 User Manual & Specs](#)
- [Alternative SSD1306 Controller Core Datasheet](#)

b. ESP32S DevKIT WIFI + BLE Module (30Pin)

- [ESP32 DevKit Technical Specifications Guide](#)
- [Ai-Thinker ESP32S Pinout & Product Page](#)

c. DHT22 AM2302 Temperature & Humidity Sensor

- [Aosong DHT22 Official Technical Datasheet](#)
- [Waveshare Wiki DHT22 Interface Guide](#)

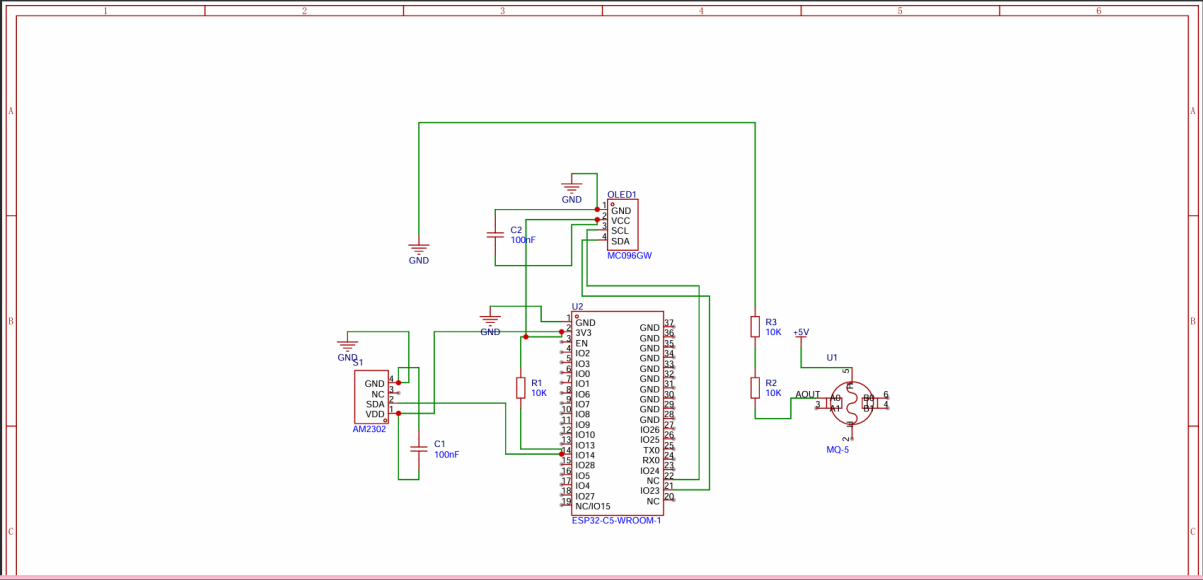
d. MQ-5 LPG, Natural Gas, & Coal Gas Sensor

- [Hanwei Electronics MQ-5 Gas Sensor Technical Datasheet](#)

e. 5V 1-Channel Low Level Trigger Relay Module

- [Standard 5V Single Channel Relay Reference Guide.](#)

4a) During schematic design in EasyEDA, a design rule check (DRC) indicated connectivity warnings (red markers) at the capacitor connections. The issue was identified but not resolved due to time constraints and will be addressed during the implementation phase. ...cause of the red dots at the capacitor points



Schematic	Schematic2			Create at	2026-06-09
				Update at	2026-06-09
Board				Page	P1
Drawn		IOT Project			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

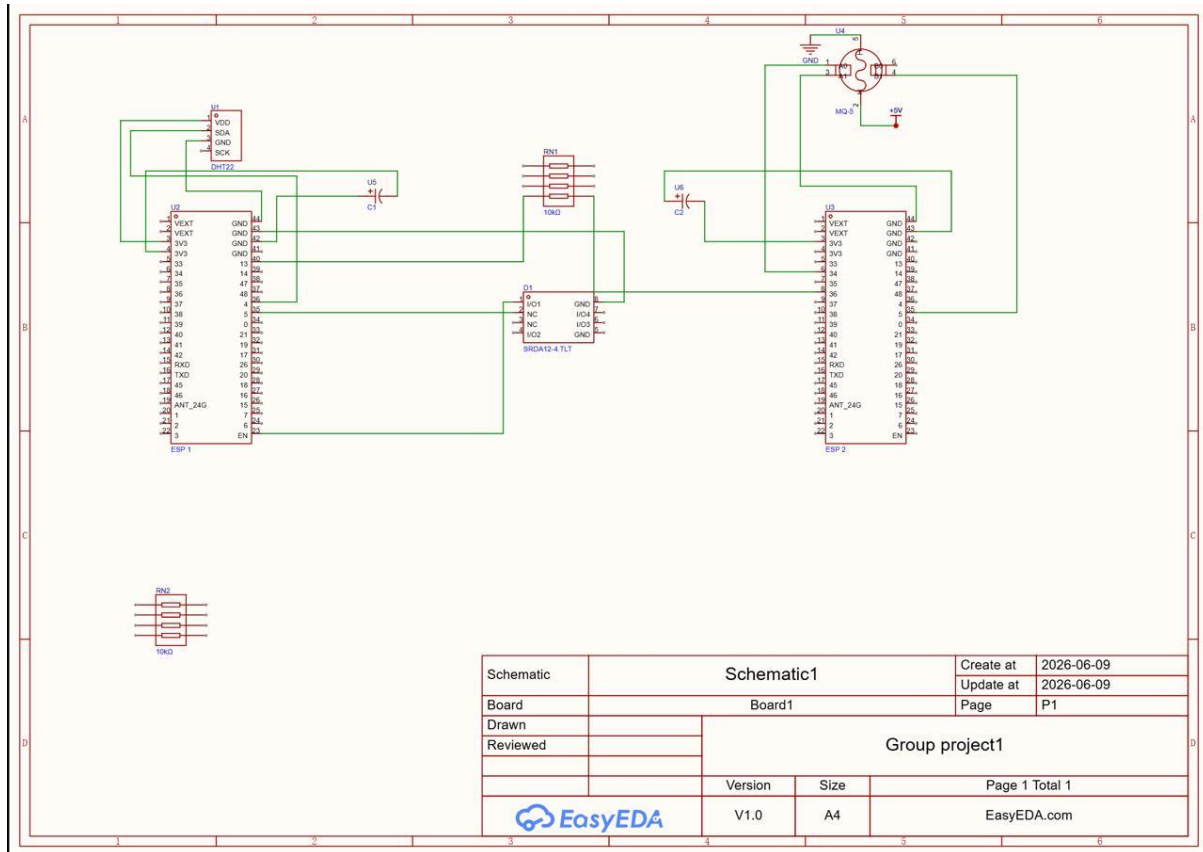
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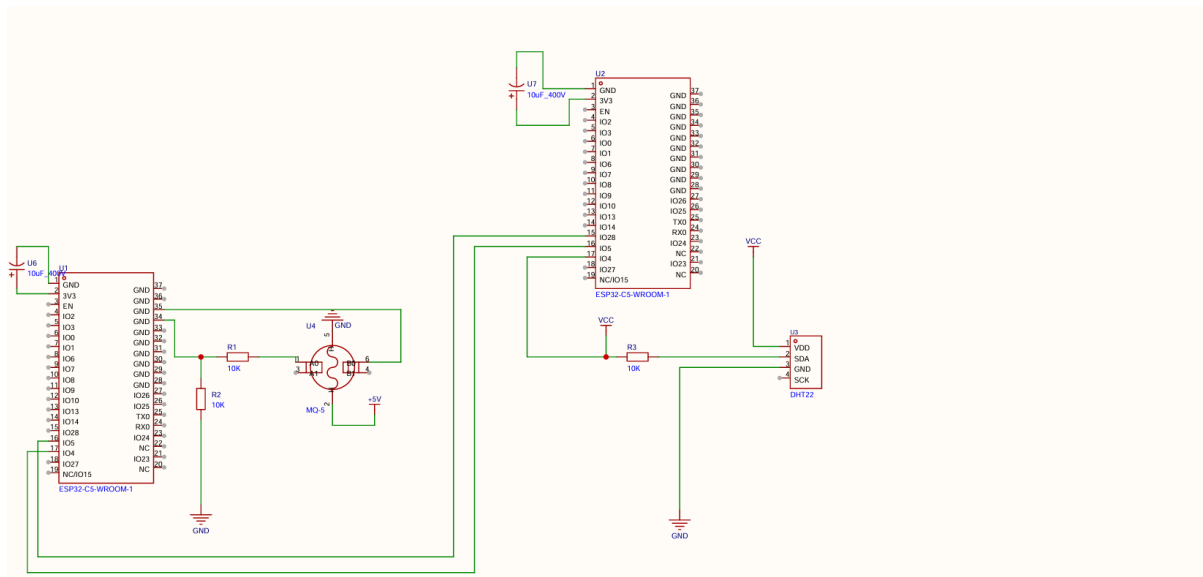
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6

4b



4c



						D	
Schematic	4b				Create at		2026-06-09
					Update at		2026-06-09
Board	Board1				Page		P1
Drawn		Deliverable1					
Reviewed							
		Version	Size	Page 1 Total 1			
		V1.0	A4	EasyEDA.com			
3		4		5			6